

Security Concepts

Module 1

Unit 4: Cyber Threats



Learning Objective

LO1

Describe cyber threats, their consequences and how they attack digital assets in an organization

LO2

Explain different types of attackers

LO3

Discuss the common motives of an attacker in a given context

LO4

Discuss different methods or approaches to conducting cyber-attacks

LO5

Describe types of cyber-attack and attack vectors using examples



TERMS FOR CYBER THREATS

Threat Source or Sponsor

is the **person** or **organization** that wishes to benefit
from attacking an **information asset**

TERMS FOR CYBER THREATS

Threat Actors or Agents

are the **individuals** or **groups** of individuals who actually
execute a cyber-attack

TERMS FOR CYBER THREATS

Threat Actions

describe the **actual attacks**. These are often **not a single isolated event**, but can consist of **many discrete activities**, involving **surveillance, initial activities, testing** and the **final attacks**.

TERMS FOR CYBER THREATS

Threat Analysis

describes the **process** of understanding the **level of threat** –
this is referred to in more detail in Chapter 6 – **Risk management
overview**

TERMS FOR CYBER THREATS

Threat Vectors or Attack Vectors

are the **tools, techniques** and **mechanisms** by which an attacker **conducts the attack** on their **target**

TERMS FOR CYBER THREATS

Threat Consequences or Impacts

are the **results** or **impacts** of a **cyber-attack**,
which we dealt with in Chapter 4

THREE THINGS NEEDED

MOTIVE

there must be **a reason** for them **undertaking a cyber-attack** – even
if it appears to be a **rather futile one**

THREE THINGS NEEDED

MEANS

the attacker must possess a minimum level of skill in order to
mount a successful attack.

Often attackers with little or no skill will fail in their endeavours and
will probably be identified and face justice

THREE THINGS NEEDED

METHOD

a more **experienced attacker** will develop **a plan** for their attack.

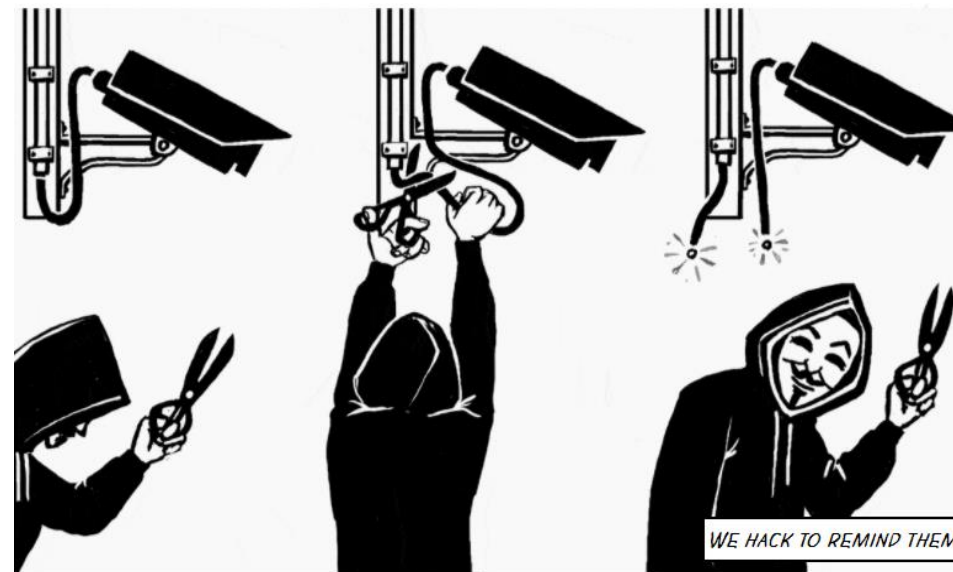
This may require an **interim break-in**, followed by **extended periods of reconnaissance** before the **real attack** takes place.

TYPES OF ATTACKER

script kiddies



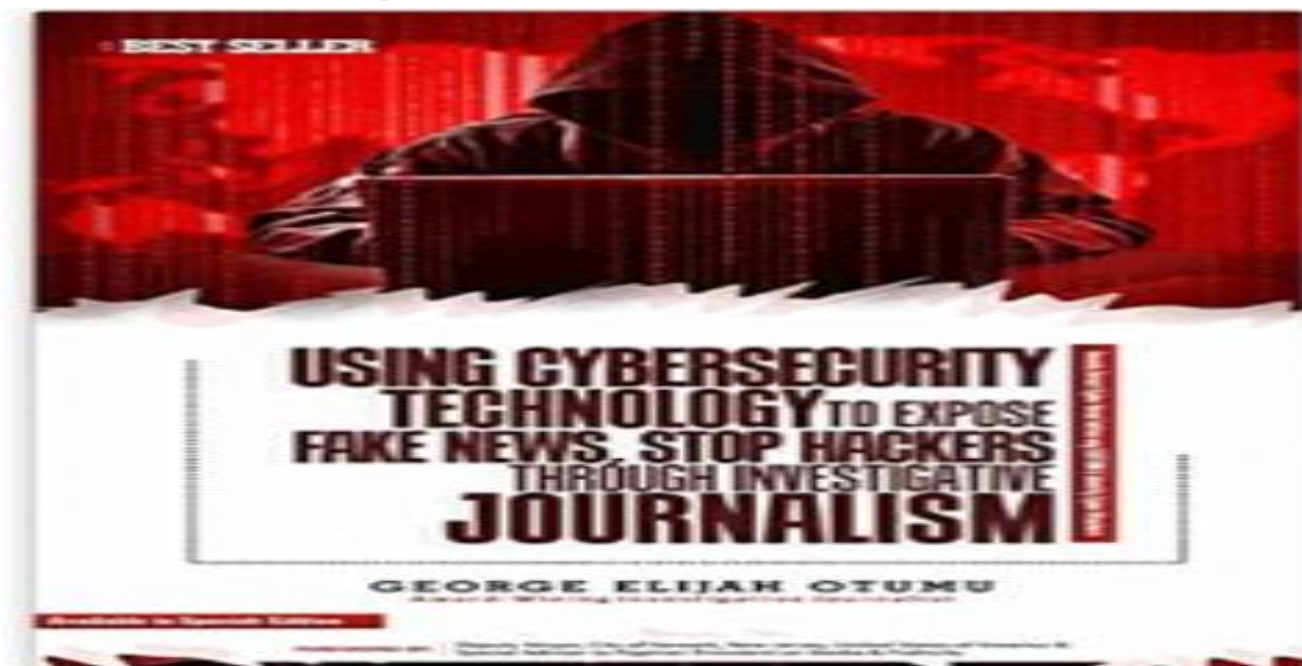
hacktivists



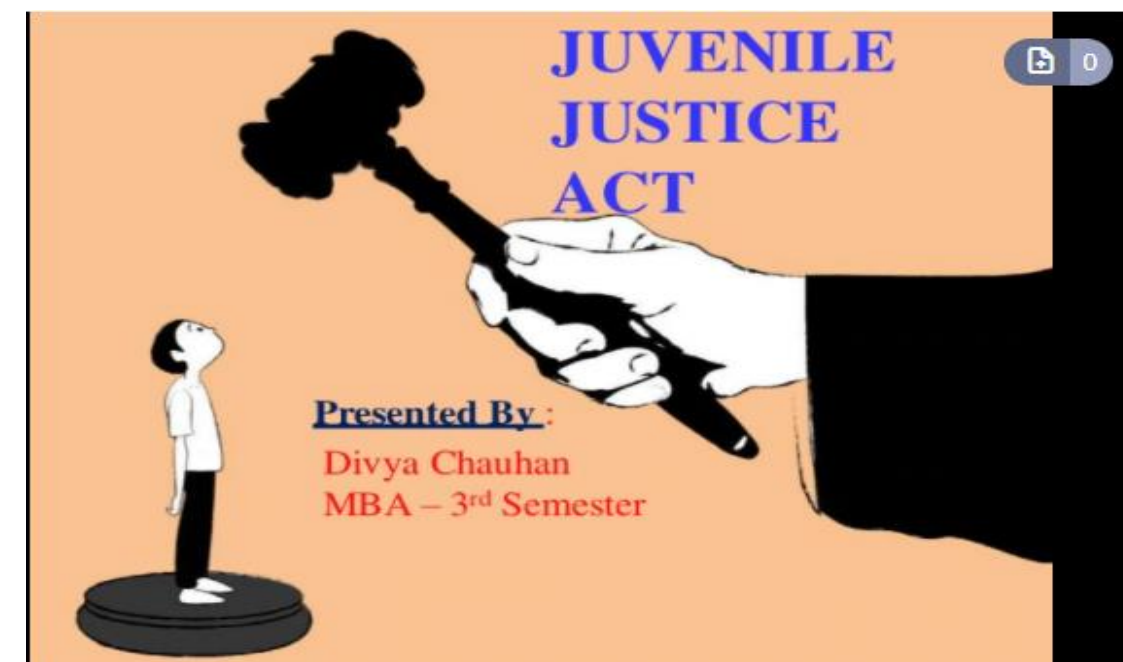
lone wolves



investigative journalists



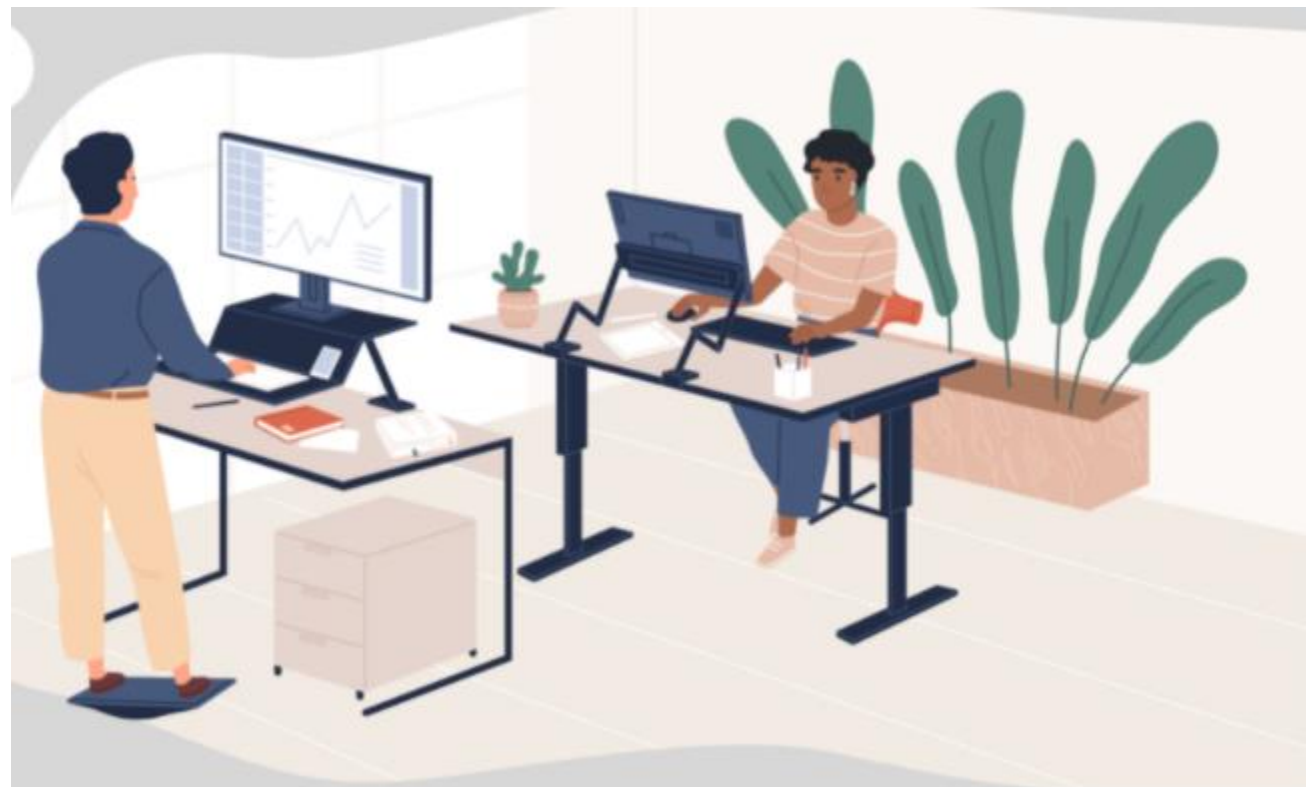
minor criminals



TYPES OF ATTACKER



organised criminals

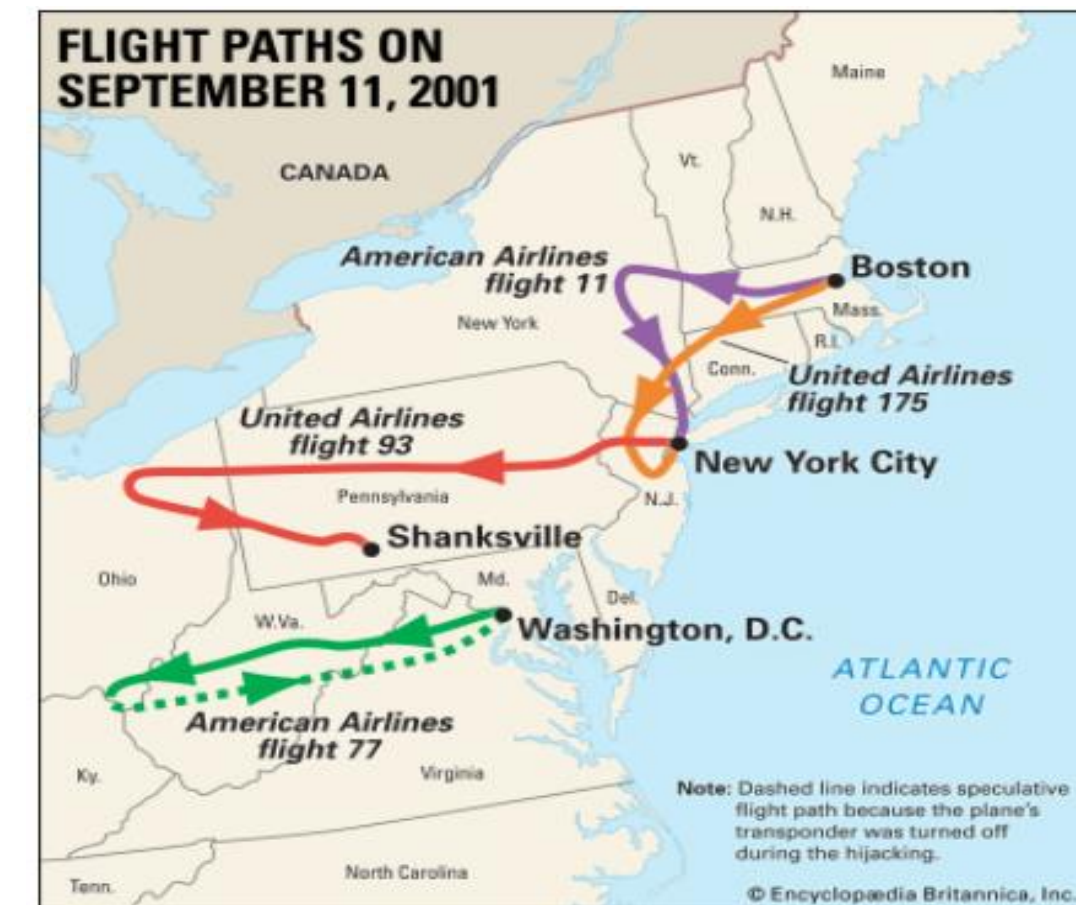


insiders



security agencies

terrorists



MOTIVES – WHAT DRIVES AN ATTACKER

Financial gain

- Financial gain generally breaks down into three distinct areas:

Ransom



Theft



Fraud



MOTIVES – WHAT DRIVES AN ATTACKER

Revenge or malicious damage

- Some cyber-attacks are carried out in response to an action undertaken or perceived to have been undertaken by the victim

Espionage



Whistle-blowing



Intellectual property theft



MEANS – HOW THEY GO ABOUT

Some General Tools

Kali Linux



John the Ripper



Nmap



Aircrack-NG



Wireshark



Nessus



MEANS – HOW THEY GO ABOUT

Some General Tools

Angry IP Scanner



Metasploit



CYBER-ATTACK METHODS

Stages of an attack

Reconnaissance



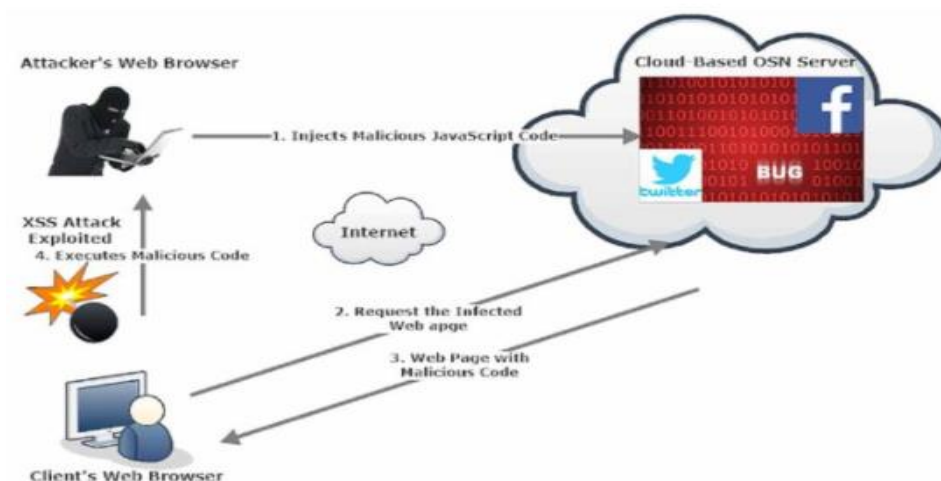
Weaponisation



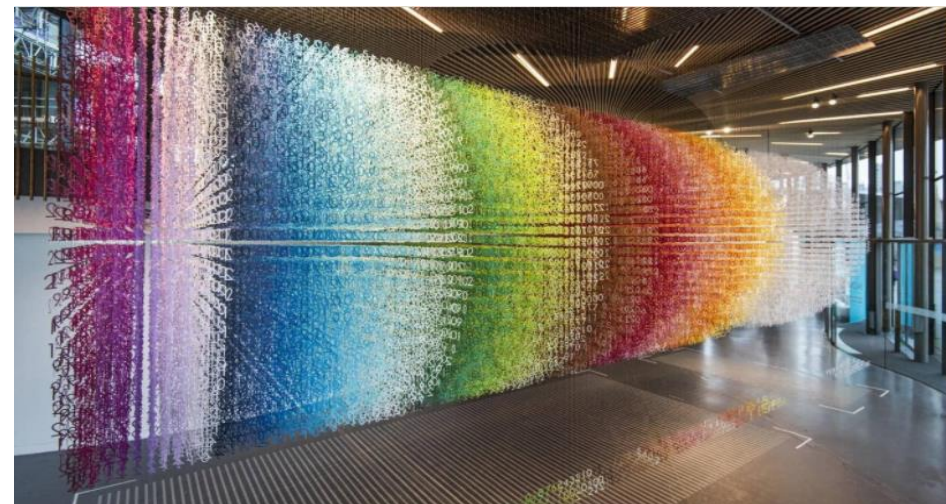
Delivery



Exploitation



Installation



Command and control



CYBER-ATTACK METHODS

Stages of an attack

Actions on objectives



PREPARE FOR A
PRODUCTIVE YEAR

